Mathematical Structure
of
Quantum Mechanics.

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10

3. $F(h,x) = e^{2hx-h^2} = e^{x^2} \cdot e^{-x^2+2hx-h^2} = e^{x^2} \cdot e^{-(x-h)^2}$

$\therefore F(h,x) = \sum_n \frac{1}{n!} \cdot h^n \cdot e^{x^2} \cdot \left[\frac{\partial^n}{\partial h^n} \cdot \left(e^{-(x-h)^2} \right) \right]_{h=0}$.

So I want to prove that $\left. \frac{\partial^n}{\partial h^n} \left(e^{-(x-h)^2} \right) \right|_{h=0} = (-1)^n \cdot \left(\frac{d}{dx} \right)^n \cdot e^{-x^2}$

By induction, for $n=1$, $\left. \frac{\partial}{\partial h} \left(e^{-(x-h)^2} \right) \right|_{h=0} = -2(x-h) \cdot (-1) \cdot e^{-(x-h)^2} \Big|_{h=0}$

$= +2x e^{-x^2}$.

$(-1) \frac{d}{dx} e^{-x^2} = 2x e^{-x^2}$

$\Rightarrow$ correct!

We assume that equality holds for $n=k$.

For $n=k+1$, $\frac{d}{dx} \left(\frac{d}{dx} \right)^n e^{-x^2}$

$= \frac{d}{dx} \left(\frac{\partial^n}{\partial h^n} \left(e^{-(x-h)^2} \right) \Big|_{h=0} \right) \cdot (-1)^n$

$= \frac{\partial^n}{\partial h^n} \left(\frac{\partial}{\partial x} e^{-(x-h)^2} \Big|_{h=0} \right) \cdot (-1)^n$

$= \frac{\partial^n}{\partial h^n} \cdot \left(-2(x-h) e^{-(x-h)^2} \right) \Big|_{h=0} \cdot (-1)^n$

$= (-1)^{n+1} \cdot \frac{\partial^n}{\partial h^n} \cdot \frac{\partial}{\partial h} e^{-(x-h)^2} \Big|_{h=0} = (-1)^{n+1} \frac{\partial^{n+1}}{\partial h^{n+1}} e^{-(x-h)^2} \Big|_{h=0}$.

$$\therefore \left.\frac{\partial^n}{\partial h^n} e^{-(x-h)^2}\right|_{h=0} = (-1)^n \left(\frac{d}{dx}\right)^n e^{-x^2}.$$

That is to say $F(h,x) = \sum_n \frac{1}{n!} h^n \cdot e^{+x^2} \cdot \left[\frac{\partial^n}{\partial h^n} e^{-(x-h)^2}\right]_{h=0}$

$$= \sum_n \frac{1}{n!} h^n \cdot e^{x^2} \cdot (-1)^n \left(\frac{d}{dx}\right)^n e^{-x^2}$$

$$= \sum_n \frac{1}{n!} h^n \cdot H_n(x)$$

$$\Rightarrow H_n(x) = (-1)^n \cdot \exp(+x^2) \left(\frac{d}{dx}\right)^n \exp(-x^2) \quad \#.$$

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2. $$\langle 0|X^2|0\rangle = \frac{\hbar}{m\omega} \langle 0|\xi^+|0\rangle$$

$$= \frac{\hbar}{2m\omega} \langle 0| a^+a^+ + a^+a + aa^+ + aa |0\rangle$$

$$= \frac{\hbar}{2m\omega} \langle 0| aa^+ |0\rangle = \frac{\hbar}{2m\omega} \langle 1|1\rangle = \frac{\hbar}{2m\omega}$$

$$\therefore \exp\left[-\frac{k^2}{2}\langle 0|X^2|0\rangle\right] = \exp\left[-\frac{k^2}{4}\cdot\frac{\hbar}{m\omega}\right].$$

$$\langle 0|e^{ikx}|0\rangle = \int_{-\infty}^{\infty} e^{ik'\xi} \cdot \frac{1}{\sqrt{\pi}} \cdot \exp(-\xi^2)\, d\xi, \quad \text{where } k' = k\cdot\sqrt{\frac{\hbar}{m\omega}}$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \exp\left[-\left(\xi - \frac{ik'}{2}\right)^2\right] d\xi \cdot \exp\left[-\frac{k'^2}{4}\right]$$

$$= \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} \cdot \exp\left[-\frac{k^2}{4}\cdot\frac{\hbar}{m\omega}\right] = \exp\left[-\frac{k^2}{4}\cdot\frac{\hbar}{m\omega}\right]$$

$$\Rightarrow \exp\left[-\frac{k^2}{2}\langle 0|X^2|0\rangle\right] = \langle 0|e^{ikx}|0\rangle \quad \#$$

1. $\langle n|\xi|n\rangle = \frac{1}{\sqrt{2}}\langle n|a^+ + a|n\rangle = 0$

10

$$\langle n|\xi^2|n\rangle = \frac{1}{2}\langle n|a^+a^+ + a^+a + aa^+ + aa|n\rangle$$

$$= \frac{1}{2}\langle n|aa^+|n\rangle = \frac{1}{2}(n+1)$$

$\therefore$ $\Delta\xi$ in n particle state $= \sqrt{\frac{n+1}{2}}$.

$$\langle n|P_\xi|n\rangle = 0$$

$$\langle n|P_\xi^2|n\rangle = -\frac{1}{2}\langle n|a^+a^+ - a^+a - aa^+ + aa|n\rangle$$

$$= \frac{1}{2}\langle n|aa^+|n\rangle = \frac{n+1}{2}$$

$\therefore$ $\Delta P_\xi = \sqrt{\frac{n+1}{2}}$

$\Rightarrow$ uncertainty $\Delta\xi\,\Delta P_\xi = \frac{n+1}{2}$ *